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Clinical Note Routing: Leakage-Free, Explainable Specialty Classification of Medical Transcriptions

MAAI 902 Natural Language Processing & LLMs ·

Assessment Task 2: NLP Solution Design ·

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Word count: approx. 3,300 words of running text in Sections 1–11 (excluding abstract, tables, figure captions and references; approx. 4,600 including tables)

Deliverable	Location
Code repository (README, commented code)	GitHub: nourhaneemamdouh-svg/MAAI-902-project-l-1113173
Working prototype (web UI)	Clinical Note Router
Executed notebook	`notebooks/01_data_understanding_and_preprocessing.ipynb` (Colab-ready: data understanding, cleaning, pre-processing) and `notebooks/clinical_note_routing.ipynb` (modelling and evaluation)
Database	Healthcare Documentation Database

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Abstract

Clinical transcriptions need to be sent to the right medical specialty before they can be used for referral, coding or audit. This project designs and builds an NLP system that gives each transcription one or more specialties and a confidence score, and sends uncertain notes to a human reviewer. The data is the MTSamples-based course dataset with 3,813 rows. An audit of the data found two serious problems. First, the `keywords` column contains the label in every row. Second, there are only 1,821 unique notes, and 1,650 of them appear under more than one specialty.

For this reason, the task was treated as multi-label classification on 1,751 unique notes and 24 cleaned specialty labels, and it was tested with leakage-free 5-fold cross-validation. The best model, TF-IDF with logistic regression, reached a macro-F1 of 0.809 ± 0.033 and placed a correct specialty first for 93.4% of notes. Its confidence scores were well calibrated (ECE 0.039).

A fine-tuned Bio_ClinicalBERT model, tested on one fold, scored lower (macro-F1 0.544 against 0.805 for logistic regression on the same fold). With a confidence threshold of 0.7, 87.5% of notes can be routed automatically with 96.7% accuracy. An ablation study showed that the evaluation method alone can change the accuracy of the same model from 0.10 to 0.81, which helps to explain why published results on this dataset disagree. The design also includes a transformer fine-tuning pipeline, clinical named-entity recognition and a web demo that runs in the browser.

1. Introduction

Writing clinical documentation takes a large part of healthcare workers' time. In a 2024 survey of 1,253 clinicians, 77% said that electronic health record (EHR) documentation made them finish work late or work from home, and 74% said that it reduced the time they could give to patients (Healthcare Innovation, 2024). Much of this documentation is free text, such as dictated consultations, operation reports, imaging reports and discharge summaries. Before these notes can be used for referral, coding or audit, someone must decide which specialty they belong to. Doing this by hand is slow and not always consistent. Simple keyword rules do not work well either, because different specialties use many of the same words and abbreviations.

Natural language processing (NLP) can help to automate this step. Several studies show that language models trained on clinical text understand medical language better than general models, and that newer models can read much longer documents (Alsentzer *et al.*, 2019; Li *et al.*, 2023; Sounack *et al.*, 2025). Large language models (LLMs) have also written clinical summaries that doctors rated as good as, or better than, summaries written by experts (Van Veen *et al.*, 2024). However, studies that classify MTSamples notes by specialty report very different results, from a macro-F1 of 0.28 to almost perfect scores (Barman and Barman, 2026; Pottipati and Polamreddy, 2026).

This report designs and tests a system that routes clinical notes to the correct specialties, and it explains why earlier results disagree. Its main contributions are:

- an audit of the data that finds label leakage, duplicated notes and mixed labels, and a cleaned multi-label dataset that fixes these problems;
- a leakage-free evaluation method and a comparison of four classical models, together with a pipeline for fine-tuning clinical transformers on the same data splits;

- a routing design that combines confidence scores, a human-review threshold, word-level explanations and clinical entity extraction;
- an ablation study that shows how much the evaluation method alone changes the results; and
- an ethical risk assessment with practical solutions, and a working prototype.

2. Problem Definition and Justification

2.1 Problem statement

Medical records teams, referral coordinators and clinical coders receive a constant flow of transcribed notes, and each note must reach the right specialty. If a note goes to the wrong place, care and coding are delayed. The problem in this project is: **given the text of a clinical transcription, assign every specialty that should receive it, show how confident the system is, and flag notes that need a human decision.**

Two features of real notes shape this problem. First, one note can belong to more than one specialty. For example, a knee arthroscopy report belongs to both Orthopedics and Surgery. Second, mistakes do not all have the same cost. A confident but wrong answer is worse than sending the note to a person to check.

2.2 Why NLP is needed

- **The input is free text.** Notes are about 400 words long on average, mix headed sections with narrative, and contain abbreviations such as *EF* and *ORIF*. Simple database queries cannot sort them.
- **Specialties share vocabulary.** Words such as *pain*, *MRI* and *biopsy* appear in many specialties. The meaning comes from word combinations and context, which language models can learn but keyword rules cannot.
- **Suitable tools already exist.** Clinical language models and biomedical entity recognition tools are freely available, so a reliable solution can be built at low cost (Alsentzer *et al.*, 2019; Neumann *et al.*, 2019; Li *et al.*, 2023).

2.3 Research gap

- **Results disagree and may be inflated.** On MTSamples, reported scores range from 0.28 macro-F1 with leakage-free testing to 0.996 F1 when oversampling is used (Barman and Barman, 2026; Pottipati and Polamreddy, 2026). Data leakage is a well-known reason why machine learning results cannot be reproduced (Kapoor and Narayanan, 2023).
- **The label list mixes two problems.** Some MTSamples labels are document types, such as *Discharge Summary*, and not medical specialties.
- **Confidence is rarely studied.** Most studies report only accuracy. A routing tool also needs reliable confidence scores to decide when a person should check a note.

2.4 Objectives and research questions

1. Audit the data and build a clean, leakage-free, multi-label version with a correct list of specialties.
2. Compare classical models and design a transformer pipeline, using macro-F1 as the main metric.
3. Add confidence scores, a human-review threshold, explanations and entity extraction.
4. Measure how much evaluation choices change the reported results.

- **RQ1:** How well can classical text classifiers route notes when there is no data leakage?
- **RQ2:** How much do the label list and the evaluation method change the reported scores?
- **RQ3:** Can confidence scores safely separate notes that can be routed automatically from notes that need human review?

2.5 Scope

The project covers English transcriptions, 24 specialty labels, a prototype for classification and entity extraction, and a demonstration. It does not cover diagnosis or treatment advice, real patient records, or connection to a hospital system.

3. Related Work

Clinical language models. Several studies agree that models trained on clinical text perform better on clinical tasks than general models (Alsentzer *et al.*, 2019; Li *et al.*, 2023; Sounack *et al.*, 2025). ClinicalBERT continued the training of BERT on MIMIC-III notes, but it can read only 512 tokens at a time (Alsentzer *et al.*, 2019). Later models removed this limit. Clinical-Longformer and Clinical-BigBird read up to 4,096 tokens, and BioClinical ModernBERT reads up to 8,192 tokens. Both outperformed ClinicalBERT on most clinical tasks, including document classification (Li *et al.*, 2023; Sounack *et al.*, 2025).

Larger or newer models are not always better. Other studies show that transformer or medically trained models do not automatically win. On medical transcriptions, Naive Bayes reached 86% accuracy while BERT-based models reached about 63% (Kathirvel and Mookiah, 2025). In large LLM benchmarks, many medically fine-tuned models performed worse than newer general models, or worse than the general models they were built from (Dada *et al.*, 2025; Wu *et al.*, 2025). Zero-shot open-source LLMs were also weaker on clinical notes than on social media text, and open models were preferred for privacy and reproducibility (Guo and Sarker, 2025). Together, these findings show that a simple baseline should always be tested.

Specialty classification on MTSamples. Results on this dataset depend strongly on how the model is tested. Pottipati and Polamreddy (2026) reported a weighted F1 below 0.20 on the original data, but 0.996 after SMOTE oversampling, and they did not describe a separate test set. Barman and Barman (2026) argued that earlier work applied SMOTE before splitting the data. With a leakage-free split of 4,966 records and 40 labels, their best macro-F1 was 0.279. This agrees with the general warning that data leakage makes results look better than they really are (Kapoor and Narayanan, 2023).

Study	Data and labels	Evaluation	Main result	Gap this project addresses
Pottipati and Polamreddy (2026)	MTSamples, 40 single labels	SMOTE, split not described	F1 0.996	Leakage; no macro-F1
Barman and Barman (2026)	MTSamples, 40 labels	Leakage-free split	Macro-F1 0.279	Document-type labels kept
Kathirvel and Mookiah (2025)	Medical transcriptions	Single-label	NB 86% acc., BERT ≈63%	Duplicates and calibration not studied
This project	1,751 unique notes, 24 labels	Multi-label, iterative-stratified 5-fold CV	Macro-F1 0.809	Audit, label fix, calibration, review

No study found in this review combines a leakage audit, a corrected label list, calibrated confidence and a human-review step. This project aims to fill that gap.

4. Data Description and Analysis

4.1 Source and structure

The dataset is the course file `Healthcare_Documentation_Database.csv`. It has 3,813 rows, 7 columns and no missing values. It comes from MTSamples, a public website of transcription samples written by users, where names and dates have been changed or removed (MTSamples, n.d.).

Column	Content	Used as model input?
`transcription`	Full note text (1–1,873 words)	**Yes**, the only input
`medical_specialty`	One of 39 labels per row	Target
`keywords`	Comma-separated terms	**No**: starts with the label in 100% of rows
`description`, `sample_name`	Short human summary and title	No: not available when a new note arrives
`cleaned_transcription`	Pre-cleaned copy	No: stop-words already removed, not suitable for transformers
`Serial No`	Row number	No

4.2 Audit findings

- **Label leakage.** Every value in the `keywords` column starts with the specialty name, for example *"urology, circumcision, …"*. A model that uses this column learns the answer instead of learning from the note.
- **Duplicated notes.** The 3,813 rows contain only 1,821 different transcriptions. Of these, 1,650 appear under two or more specialties, with an average of 2.09 labels per note. This is not an error: a colonoscopy report really belongs to both Gastroenterology and Surgery. Therefore, the task is **multi-label**.
- **Mixed labels.** Seven labels describe the type of document, not a specialty: *Consult – History and Phy.*, *SOAP / Chart / Progress Notes*, *Discharge Summary*, *Office Notes*, *Letters*, *Emergency Room Reports* and *IME-QME-Work Comp etc.*

4.3 Exploratory analysis of the cleaned data

After cleaning (Section 5), 1,751 notes and 24 specialties remain, with an average of 1.82 labels per note.

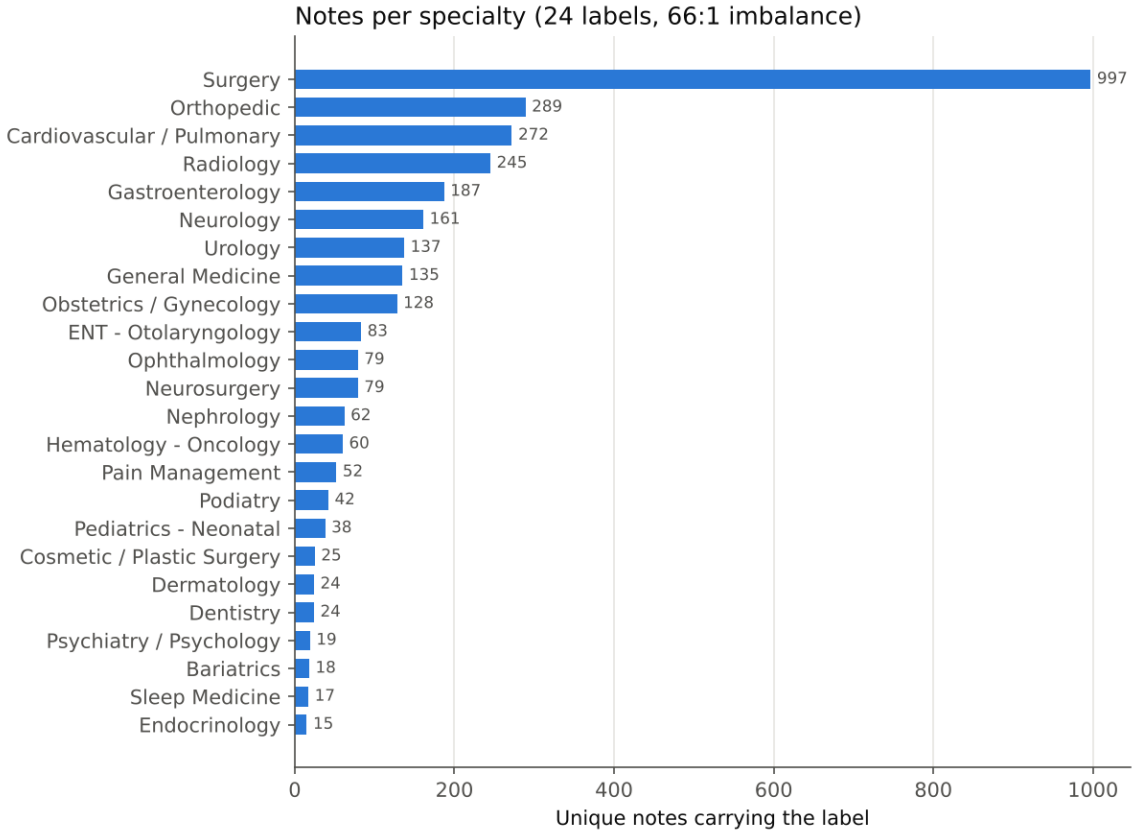


Figure 1. Surgery appears on 997 notes and Endocrinology on only 15, an imbalance of 66 to 1. For this reason, macro-F1 is used as the main metric.

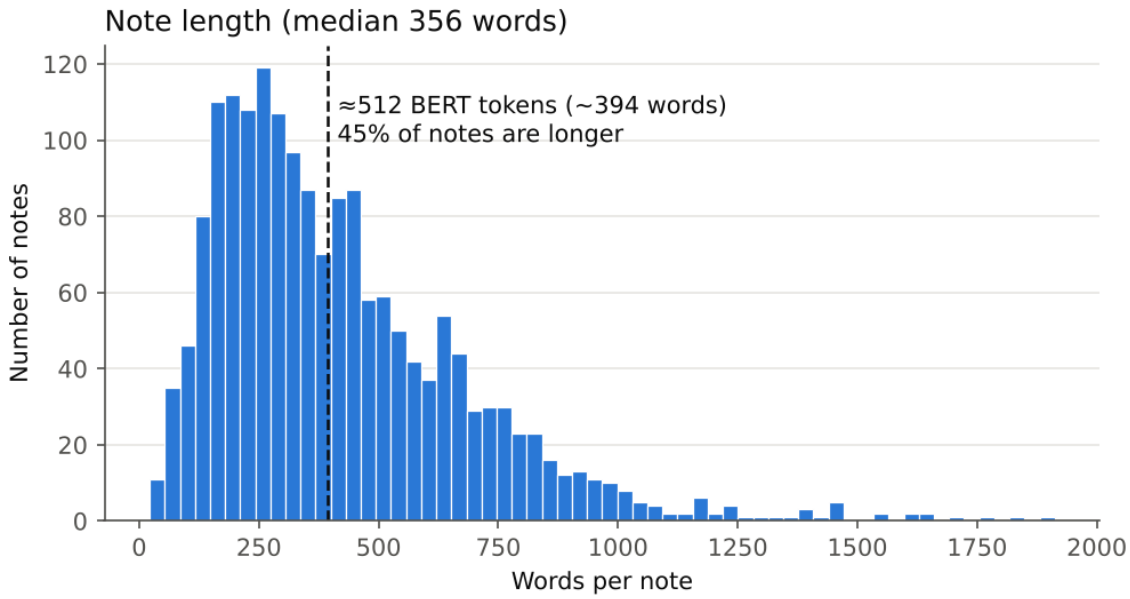


Figure 2. The median note has 356 words and the 90th percentile has 765 words. About 45% of notes are longer than about 512 BERT tokens (estimated at 1.3 tokens per word), so a standard BERT model would cut them short.

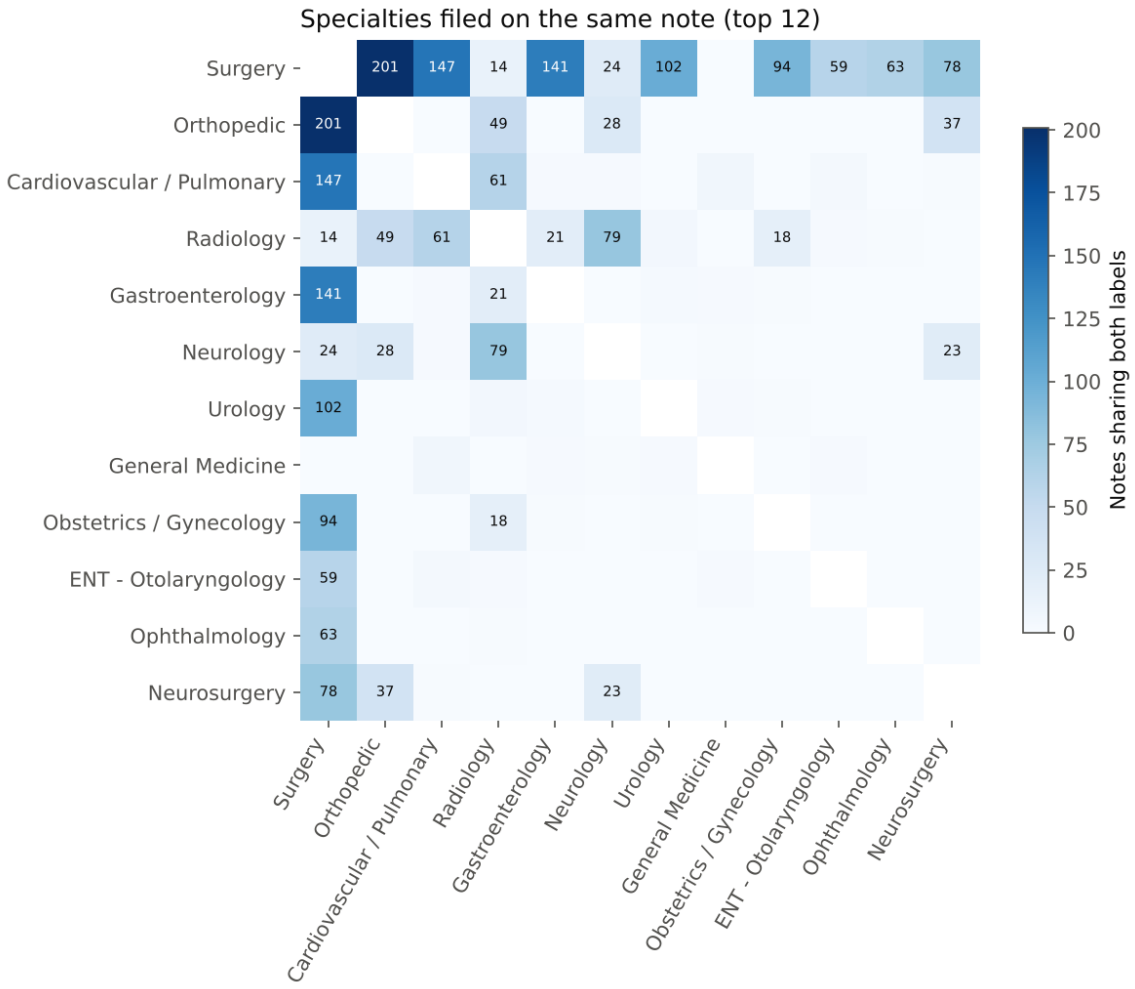


Figure 3. Surgery often appears together with procedure-based specialties (201 notes with Orthopedic and 147 with Cardiovascular / Pulmonary), and Radiology often appears with Neurology (79 notes). This overlap is part of how the data is organised, not an accident.

5. Pre-processing and Text Representation

5.1 Cleaning steps

Each step solves a problem found in Section 4. The code is in `src/data_prep.py`, and the notebook `01_data_understanding_and_preprocessing.ipynb` shows every step with before-and-after examples.

1. **Remove leaky and unavailable columns** (`keywords`, `description`, `sample_name`), so the model uses only the note text.
2. **Normalise the text:** change it to lower case, replace punctuation and the MTSamples line-break marks (`.,`) with spaces, and remove extra spaces. Stop-words are kept in the text for transformers and are removed only inside the TF-IDF step.

3. **Remove duplicates** using the cleaned text (3,813 rows become 1,821 notes) and combine the labels of each note into one set.
4. **Fix the label list:** remove the 7 document-type labels and the 8 specialties with fewer than 15 unique notes (for example, Allergy / Immunology has only 3). With so few notes, each test fold would contain two examples or fewer.
5. **Remove unusable notes:** 47 notes with no specialty left and 23 notes with fewer than 20 words.

The final dataset has 1,751 notes, 24 labels and a binary label matrix.

5.2 Leakage-free splitting

The data is split using unique notes, so the same text cannot appear in both the training and the test data. Normal stratified K-fold works only when each item has one label, so this project uses **iterative stratification** (Sechidis, Tsoumakas and Vlahavas, 2011). This method spreads every label evenly across the folds, starting with the rarest label. Each of the five folds contains between 347 and 354 notes. All thresholds and settings are tuned on the training data only.

5.3 Class imbalance

Class imbalance is handled in the loss function instead of by creating synthetic data. Linear models use `class_weight='balanced'`, and the transformer uses a positive-class weight in its binary cross-entropy loss. Oversampling was not used because, if it is done before splitting, it copies test notes into the training data (Section 8.5). Even when it is done correctly, it adds little new information for text.

5.4 Text representations

Representation	How it works	Strengths	Limitations	Role
TF-IDF, single words (tested against single words + word pairs)	Weighted word counts; sublinear TF, `min_df` 2, English stop-words removed; 11,017 features	Fast, transparent and strong on small data; every weight can be inspected	Ignores word order and synonyms	Main model and demo
Contextual embeddings (Bio_ClinicalBERT)	WordPiece tokens passed through a transformer pre-trained on clinical notes	Understands context, abbreviations and negation	512-token limit; needs a GPU; harder to explain	Fine-tuned challenger
Long-context embeddings (Clinical-Longformer, 4,096 tokens)	Sparse attention over the whole note	Long notes are not cut short	Needs more memory	Option for long notes

For 512-token models, each long note is split into overlapping windows of 512 tokens with a stride of 128. During training, every window receives the labels of its note. At prediction time, the probabilities of all windows are averaged, so the 45% of notes that are longer than the limit are still read in full.

6. Solution Architecture and Model Design

6.1 Architecture

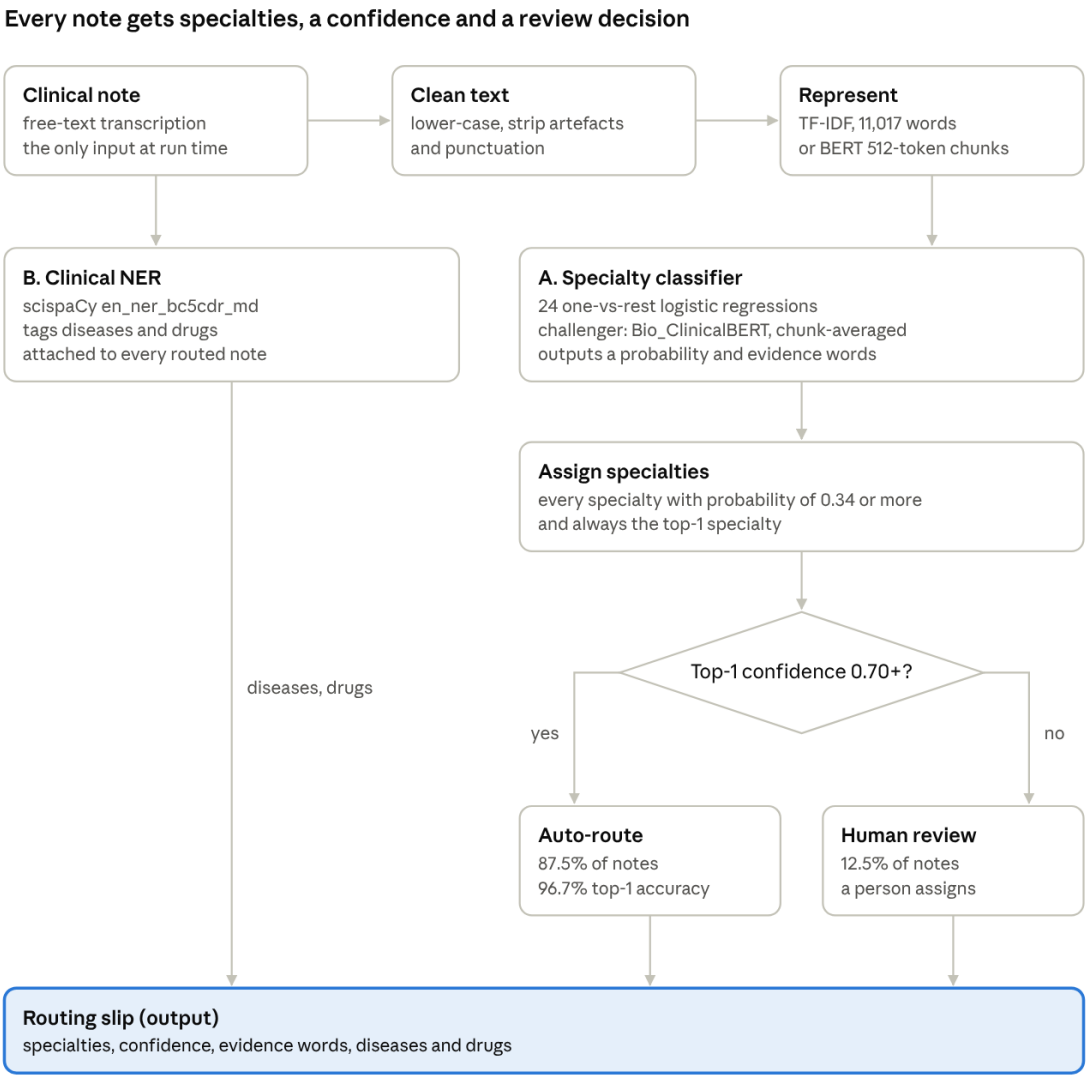


Figure 4. Solution architecture · two components, one review decision

The classifier (A) and the entity extractor (B) both read the same note. A threshold rule turns the classifier's probabilities into specialties, and a confidence check decides whether the note is routed automatically or sent for human review.

6.2 Component A: specialty classifier

The task is split into 24 binary problems (one-vs-rest), one for each specialty, so a note can receive any combination of labels. Four types of model were compared on the same folds:

Model	Why it was included
Prior (always the most frequent label)	Minimum baseline: any useful model must beat it
TF-IDF + Multinomial Naive Bayes	Classic text baseline; the best model in Kathirvel and Mookiah (2025)
TF-IDF + linear SVM (class-weighted)	Strong linear classifier for sparse text
TF-IDF + logistic regression (class-weighted)	Gives **probabilities**, which the review step needs, and a weight for every word
Bio_ClinicalBERT, chunked (challenger)	Clinical pre-training helps with context and abbreviations (Alsentzer *et al.*, 2019)

Logistic regression was chosen because it gave the best results (Section 8). It is also the only model in the comparison that is accurate, gives well-calibrated probabilities and is easy to explain.

6.3 Decision rule and human review

A note receives every specialty whose probability is above a threshold, plus its top-1 specialty, so every note is routed somewhere. The threshold is tuned for micro-F1 on 15% of the training data in each fold (0.34 for the final model). A second threshold, τ, is applied to the top-1 confidence, and notes below τ are sent to a person. This idea is called **selective prediction**, and it works well only when the confidence scores are calibrated (Guo *et al.*, 2017; Geifman and El-Yaniv, 2017). The default value is $\tau = 0.7$, and users can change it in the prototype.

6.4 Explainability

In a linear model, the contribution of each word to the top specialty is its TF-IDF value multiplied by the model weight. The words with the highest contributions are shown as evidence and highlighted in the note. For a linear model this explanation is exact, unlike methods that only approximate the model afterwards.

6.5 Component B: clinical NER

The entity extractor uses scispaCy's `en_ner_bc5cdr_md` model, which finds disease and chemical (drug) names (Neumann *et al.*, 2019). Because this model was built for an older version of spaCy, the code switches automatically to a Hugging Face biomedical NER model (`d4data/biomedical-ner-all`) when scispaCy cannot be loaded, for example on Python 3.13. The most frequent diseases and drugs are added to the routing slip, so the receiving team can see the main problems and medicines quickly. NER only gives extra information and does not affect the routing decision.

6.6 Training and tuning

Setting	Classical model	Transformer (`train_transformer.py`)
Search	Grid: C ∈ {1, 4, 16, 32, 64} × n-grams {1, 1–2}, 5-fold CV	Fixed settings, early stopping on validation loss

Setting	Classical model	Transformer (`train_transformer.py`)
	macro-F1	
Selected	C = 64, single words (CV macro-F1 0.809)	Learning rate 2e-5, 4 epochs, batch 8 × 2 accumulation, warm-up 10%, weight decay 0.01
Imbalance	`class_weight='balanced'`	BCE with positive-class weight (negatives/positives, maximum 20)
Long notes	Whole note	512-token windows, stride 128, averaged

Macro-F1 stopped improving after C = 16 (0.803, 0.805 and 0.809 for C = 16, 32 and 64). These differences are much smaller than the standard deviation between folds (0.033), so the search was not extended. Single words performed better than single words plus word pairs at every value of C (best result with word pairs: 0.787), probably because word pairs double the number of features while there are only 1,751 notes.

Alternatives that were rejected: zero-shot LLM prompting, because sending notes to an external service raises privacy concerns and zero-shot LLMs perform worse on clinical notes (Guo and Sarker, 2025); SMOTE or random oversampling, because of the leakage risk (Section 8.5); and a single-label approach, because the data clearly shows that many notes have several labels (Section 4.2).

7. Tool Stack and Platforms

Tool	Role in the project	Why it was chosen
pandas, NumPy	Loading, checking and de-duplicating the data; building label sets	Standard and fast tools for tables
scikit-learn (Pedregosa *et al.*, 2011)	TF-IDF, linear models, metrics and threshold tuning	Mature and reproducible; gives probabilities and weights that can be inspected
Custom iterative stratification	Creating the multi-label folds	Implements Sechidis, Tsoumakas and Vlahavas (2011) in about 30 lines without extra libraries
Hugging Face Transformers (Wolf *et al.*, 2020)	Fine-tuning Bio_ClinicalBERT and Clinical-Longformer; NER fallback model	Easy access to pre-trained clinical models and a tested training loop
scispaCy (Neumann *et al.*, 2019)	Disease and drug recognition	Biomedical models that run locally and quickly

Tool	Role in the project	Why it was chosen
matplotlib	Figures 1–3 and 5–7	Reproducible figures with a colour-blind-safe palette
Gradio	Python web interface (`src/app.py`)	Builds a shareable interface with little code
Plain JavaScript	Browser demo (`demo/`)	The model runs on the user's device, so no clinical text is sent anywhere
Google Colab	Running the two notebooks; GPU for transformer training	Free and easy to share; a T4 GPU is enough for a 110M-parameter model
GitHub	Version control and README	Required for submission; displays the notebooks and diagram

LangChain was considered for a feature that retrieves similar past cases, but it was not used. Routing does not need text generation, and a retrieval system would add complexity without a measured benefit for this task. The JavaScript version of the model gives the same top-1 prediction as scikit-learn on 25 test notes, and the probabilities differ by less than 0.0001 (`demo/check_parity.js`).

8. Evaluation and Results

8.1 Metrics

- **Macro-F1 (main metric):** the average F1 across the 24 specialties, so a small specialty such as Endocrinology counts as much as Surgery.
- **Micro-F1:** F1 calculated over all note–label decisions together. It shows how well the system handles the total volume of notes.
- **Top-1 routing accuracy:** the percentage of notes whose highest-ranked specialty is one of the correct ones. This is what a routing queue experiences in practice.
- **Expected calibration error (ECE):** the gap between the model's confidence and its real accuracy (Guo *et al.*, 2017). It shows whether the review threshold can be trusted.

All results are the mean ± standard deviation across the five folds, measured on notes that the model did not see during training.

8.2 Model comparison (RQ1)

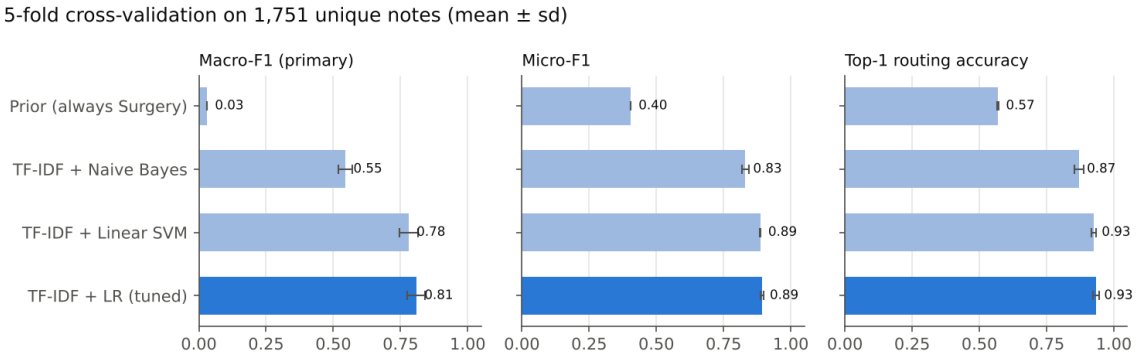


Figure 5. Results of 5-fold cross-validation on 1,751 unique notes. Logistic regression is the best model on every metric.

Model	Macro-F1	Micro-F1	Top-1 accuracy
Prior (always Surgery)	0.030 ± 0.000	0.404 ± 0.001	0.569 ± 0.004
TF-IDF + Naive Bayes	0.545 ± 0.025	0.830 ± 0.013	0.870 ± 0.018
TF-IDF + linear SVM	0.780 ± 0.035	0.886 ± 0.002	0.925 ± 0.009
TF-IDF + logistic regression (tuned)	**0.809 ± 0.033**	**0.892 ± 0.006**	**0.934 ± 0.011**
Bio_ClinicalBERT, chunked (fold 0 only)	0.544	0.741	0.831
TF-IDF + logistic regression (fold 0 only, for comparison)	0.805	0.888	0.931

Naive Bayes has a much lower macro-F1 (0.545) because its probabilities are poorly scaled for rare labels. The SVM is close to logistic regression, but it gives margins instead of probabilities.

Bio_ClinicalBERT was fine-tuned on a Google Colab T4 GPU for one fold (fold 0), which took about 5.5 minutes of training. On the same fold, logistic regression performed much better (macro-F1 0.805 against 0.544, and top-1 accuracy 93.1% against 83.1%). This agrees with earlier findings that transformer and medical models do not always outperform simpler models, especially on small datasets (Kathirvel and Mookiah, 2025; Dada *et al.*, 2025). There are three likely reasons. First, each fold has only about 1,200 training notes for 24 labels, which is little data for a 110-million-parameter model. Second, every 512-token window inherits all the labels of its note, even when that part of the note does not mention the specialty, which adds label noise. Third, the model was trained for only four epochs with default settings and no tuning. Because only one fold was run, this is an indicative result and not a final comparison.

8.3 Error analysis

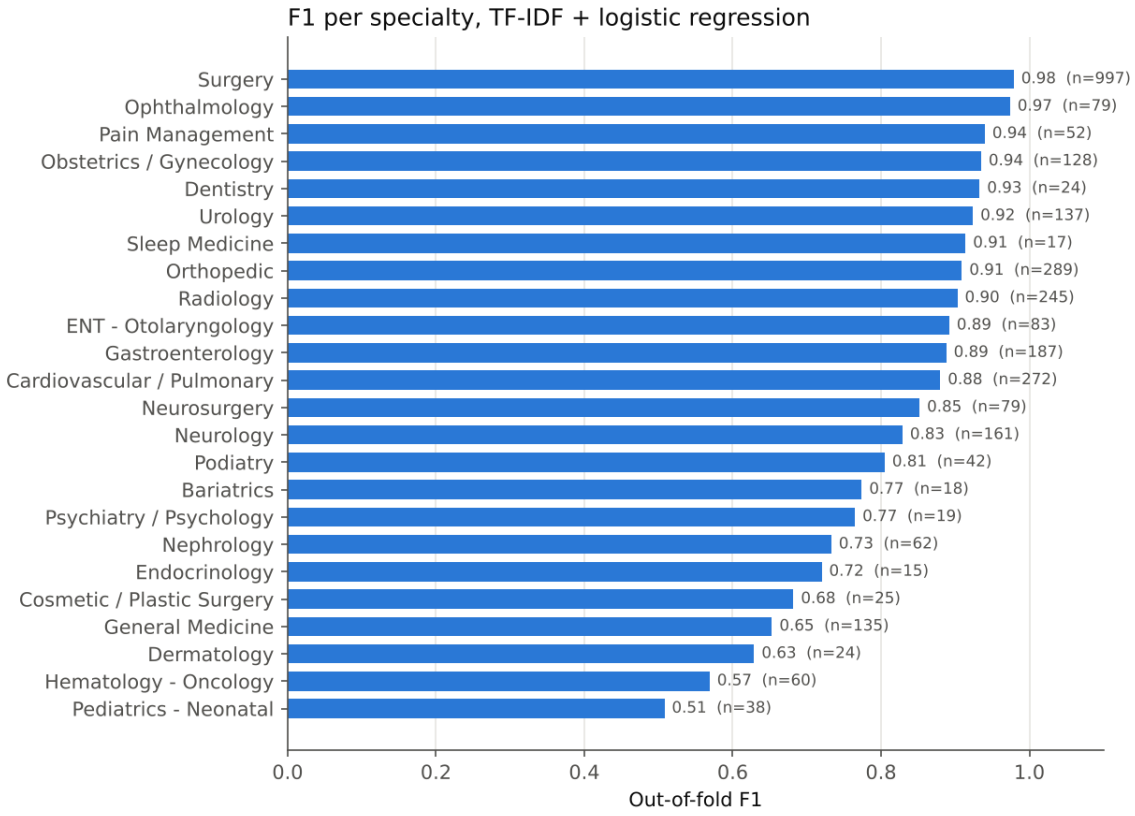


Figure 6. F1 for each specialty for the selected model, measured on the test folds.

Specialties that are defined by procedures score highest (Surgery 0.98, Ophthalmology 0.97, Obstetrics / Gynecology 0.94). The weakest specialties are defined by the *patient group* or by a *broad scope*, not by special vocabulary: Pediatrics – Neonatal 0.51, Hematology – Oncology 0.57, Dermatology 0.63 and General Medicine 0.65. Small classes are not always weak. Sleep Medicine has only 17 notes but reaches 0.91, because its words (*sleep*, *apnea*, *electroencephalogram*) are very specific. The words with the highest model weights also make clinical sense (Urology: *bladder, prostate*; ENT: *ear, nasal, tympanic*), which increases trust in what the model has learned. Radiology depends on report words such as *exam, images, impression*, which shows that it is partly a document style.

8.4 Calibration and human review (RQ3)

The top-1 confidence is well calibrated (ECE 0.039), so it can be used to decide which notes need review.

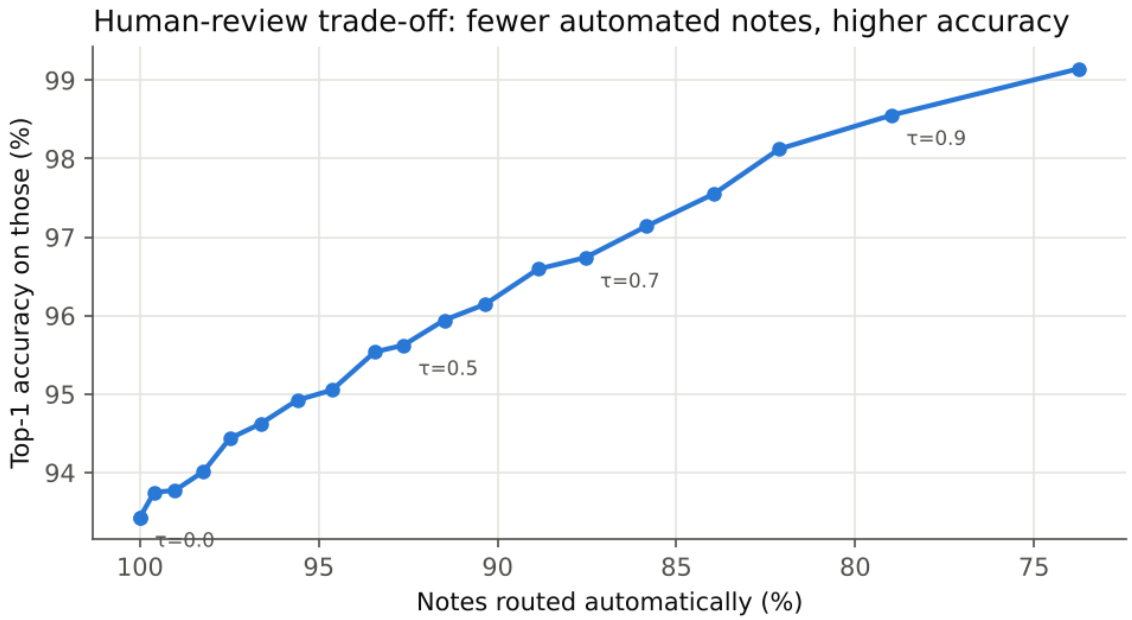


Figure 7. A higher review threshold τ sends more notes to people and increases accuracy on the remaining notes.

Review threshold τ	Routed automatically	Top-1 accuracy on those
none	100%	93.4%
0.5	92.6%	95.6%
0.7 (default)	87.5%	96.7%
0.9	79.0%	98.6%

This means a hospital can choose its own balance. At τ = 0.7, one note in eight is checked by a person, and the error rate on the automatically routed notes is halved, from 6.6% to 3.3%.

8.5 Ablation: the evaluation method changes the score (RQ2)

The same TF-IDF and logistic regression model was tested with the different methods used in earlier work (`src/leakage_experiments.py`).

#	Evaluation method	Accuracy	Macro-F1
E0	Raw rows, one label each, random split	0.104	0.057
E1	E0 + oversampling **before** the split	0.807	0.783
E2	E0 + oversampling after the split	0.104	0.071
E3	E2 + `keywords` column	0.722	0.529
E4	Raw rows, split grouped by text	0.444	0.202

#	Evaluation method	Accuracy	Macro-F1
E5	Multi-label, unique notes, all 39 labels	top-1 0.946	0.695
E6	Multi-label, unique notes, 24 cleaned labels (this design)	top-1 0.934	0.809

The results lead to three conclusions:

- **The single-label approach does not work for this data.** In E0, the text of a test note is usually also in the training data, but with a different label that is also correct. The model predicts that other label and is marked wrong, so accuracy is only 0.10. This is even lower than with a grouped split (E4, 0.44).
- **Leakage makes results look much better than they are.** Oversampling before splitting (E1) and using the `keywords` column (E3) raise accuracy to 0.81 and 0.72 without any real improvement in the model. This supports the warnings about leakage in earlier work (Kapoor and Narayanan, 2023; Barman and Barman, 2026), and it may explain the near-perfect score reported by Pottipati and Polamreddy (2026).
- **Fixing the label list adds real value.** Moving from 39 to 24 labels (E5 to E6) raises macro-F1 from 0.695 to 0.809. Part of this increase comes from removing rare and difficult labels, so the two numbers measure slightly different tasks, and both are reported.

9. Ethical and Responsible AI Considerations

A routing system does not make diagnoses, but it decides which clinician sees a note and when. Its errors and biases can therefore affect patients.

Risk	Possible impact	How this design reduces it
Patient privacy	Clinical notes reveal identity and health information; sending them to external services may break HIPAA or GDPR rules	Only de-identified public data is used; the model runs locally and the browser demo sends no text anywhere; no external LLM services are used; a real system would add automatic de-identification, local hosting and access logs
Bias between specialties	Specialties defined by patient group, such as Pediatrics (F1 0.51), are served worse, so their notes may be delayed	F1 is reported for every specialty, not only the average; the loss is class-weighted; low-confidence notes go to review; results per specialty would be monitored after deployment
Automation bias	Staff may trust a confident but wrong decision without checking it	Every routing slip shows the confidence; the review threshold is clear and adjustable; evidence words let a reviewer check the reason in seconds
Overstating performance	Inflated results can lead to deployment before the system is ready	Data audit for leakage, folds built on unique notes, an ablation of evaluation methods, and variation across folds reported

Risk	Possible impact	How this design reduces it
Lack of transparency	Users cannot question a decision they do not understand	Exact word-level explanations from the linear model; a short model summary in the prototype
Data source and consent	MTSamples notes are written by users and are "not guaranteed to be complete or error free" (MTSamples, n.d.)	Used only as a research example; testing on approved hospital data with patient consent is needed before any real use
Hallucination (if LLMs are added)	Generated summaries can include invented findings (Van Veen *et al.*, 2024)	The current design has no text generation; any future summary feature would be based only on the source note and checked by a person
Accountability	It may be unclear who is responsible when a note is misrouted	The system supports people and does not decide alone; people are responsible for every reviewed note, and logs record the model version, confidence and decision

The human-review threshold is the most important safety feature. Because the confidence scores are calibrated (ECE 0.039), a hospital can choose τ based on a clear trade-off instead of a guess. It can also use a stricter threshold for sensitive queues such as Pediatrics.

10. Expected Output and Value Proposition

10.1 What the user receives

For every note, the system produces a routing slip. The example below is a lumbar spine CT report from the dataset, taken from the prototype:

Field	Output
Decision	Route automatically (top-1 confidence 99.96%, above τ = 0.70)
Specialties assigned	Radiology, Orthopedic, Neurology
Labels in the source data	Neurology, Orthopedic, Radiology
Evidence words	cyst, images, spine, mild, exam, impression, technique, demonstrate
Entities (when NER is available)	Diseases and drugs mentioned in the note

10.2 Prototype

In the Clinical Note Router demo, a user can choose one of eight sample notes or paste their own text. The demo shows the routing decision, the assigned specialties, a probability bar for each specialty with the threshold marked, the supporting words highlighted in the note, and a slider to change the review threshold. The same model is also available as a Gradio app (`src/app.py`), which also lists the diseases and drugs found by NER, and in two Colab notebooks.

10.3 Value

- **Medical records and referral teams:** about seven out of eight notes are routed without manual reading, and the uncertain notes are clearly marked.
- **Clinical coders and auditors:** consistent specialty labels and a list of diseases and drugs make coding and auditing faster.
- **The hospital:** a small, transparent model that runs on a normal laptop in seconds, needs no GPU or external service, and can be checked word by word.
- **Researchers:** a reproducible and leakage-free way to evaluate models on MTSamples, so that future results can be compared fairly.

11. Limitations, Future Work and Conclusion

11.1 Limitations

- **Data:** the project uses 1,751 de-identified sample transcriptions, not real hospital records. Real notes are usually longer, less tidy and different from one hospital to another.
- **Labels:** MTSamples labels show how users filed the samples; they were not assigned by medical experts. The 15 removed labels are not routed by the system.
- **Transformer tested on one fold only:** Bio_ClinicalBERT was run on fold 0 with default settings and no tuning, so a five-fold run with tuned settings is needed before a final comparison.
- **NER was not evaluated:** the entity extraction uses ready-made models, and their accuracy on these notes was not measured.
- **Language:** the system works only in English and uses a fixed vocabulary of 11,017 words.

11.2 Future work

1. Run Bio_ClinicalBERT on all five folds with tuned settings (more epochs and a learning-rate search), and test Clinical-Longformer, which reads whole notes without splitting them into windows.
2. Model the links between labels, for example Surgery appearing together with procedure-based specialties, as label-graph methods do (Barman and Barman, 2026).
3. Test the system on approved hospital data, with clinicians reviewing a sample of its decisions.
4. Add a separate classifier for the removed document-type labels and a feature that shows similar past cases.

11.3 Conclusion

This project designed an NLP system that routes clinical transcriptions to specialties, gives calibrated confidence scores and includes a human-review safeguard. The answers to the research questions are:

- **RQ1:** with leakage-free evaluation, a tuned TF-IDF and logistic regression model reaches a macro-F1 of 0.809 ± 0.033 and a top-1 routing accuracy of 93.4%. A first Bio_ClinicalBERT run on one fold scored lower (macro-F1 0.544 against 0.805 for logistic regression on the same fold), so the simpler model remains the better choice for this dataset.
- **RQ2:** the evaluation method alone changes the accuracy of the same model from 0.10 to 0.81, and fixing the label list raises macro-F1 from 0.695 to 0.809. This explains much of the disagreement between earlier studies.
- **RQ3:** the confidence scores are well calibrated (ECE 0.039), and a threshold of 0.7 routes 87.5% of notes automatically with 96.7% accuracy.

The main lesson of this project is that the most important decisions were about the data, not the model. Checking for leakage, treating the task as multi-label and separating specialties from document types mattered more than the choice of model.

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